



JKBOSE 2026

CLASS 11TH

COMPLEX NUMBER

MATHS

LECTURE-01

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TOPICS TO BE COVERED

1

What is a Complex Number?

2

Related Problems

INTRODUCTION



* Natural numbers:

Counting numbers are known as Natural numbers.

1, 2, 3, 4, 5, 6, ...

Q) Does the equation $x+1=1$ have a solution in $\mathbb{N}$?

Ans) It has no solution in $\mathbb{N}$.

$$\begin{aligned} x+1 &= 1 \\ \Rightarrow x &= 0 \notin \mathbb{N} \end{aligned}$$

* Whole numbers: 0, 1, 2, 3, 4, 5, 6, ...

Q) Does the equation $x+1=0$ have a solution in $\mathbb{W}$?

Ans) It has no solution in $\mathbb{W}$.

$$\begin{aligned} x+1 &= 0 \\ \Rightarrow x &= -1 \notin \mathbb{W} \end{aligned}$$

* Integers: 0, 1, -1, 2, -2, 3, -3, 4, -4, ...

Q) Does the equation $2x+1=0$ have a solution in $\mathbb{Z}$?

Ans) It has no solution in $\mathbb{Z}$.

$$\begin{aligned} 2x+1 &= 0 \\ \Rightarrow x &= -\frac{1}{2} \notin \mathbb{Z} \end{aligned}$$

* Rational numbers:

$$\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z} \text{ \& } q \neq 0 \right\}$$

Q) Does the equation $x^2 - 2 = 0$ have a solution in $\mathbb{Q}$?

Ans) It has no solution in $\mathbb{Q}$

$$x^2 - 2 = 0$$

$$\Rightarrow x = \pm\sqrt{2} \notin \mathbb{Q}$$

* Irrational numbers:

not rational

Real number: (Rational numbers) + (Irrational numbers)

Q) Does the equation $x^2 + 1 = 0$ have a solution in $\mathbb{R}$?

Ans) It has no solution in $\mathbb{R}$

$$x^2 + 1 = 0.$$

$$\Rightarrow x^2 = -1$$

$\Rightarrow$ there is no such $x \in \mathbb{R}$ as x^2 cannot be negative.

* Imaginary numbers:

A number x is said to be an Imaginary number if x^2 is negative.

$$* (\sqrt{-2})^2 = -2, (\sqrt{-3})^2 = -3, (\sqrt{-1})^2 = -1$$

$\Rightarrow \sqrt{-1}, \sqrt{-2}, \sqrt{-3}$ are all Imaginary numbers

Here $\sqrt{-1} = i$ (iota), So $\sqrt{-2} = \sqrt{2 \times -1}$

$$\Rightarrow \sqrt{-2} = \sqrt{2} \times \sqrt{-1} = \sqrt{2}i$$

$$\Rightarrow \sqrt{-2} = \sqrt{2}i, \sqrt{-3} = \sqrt{3}i$$

$$\text{Here } x^2 + 1 = 0.$$

$$\Rightarrow x^2 = -1$$

$$\Rightarrow x = i \text{ as } i^2 = \sqrt{-1}^2 = -1.$$

Which is an Imaginary number.

Power of i :

$$i^0 = 1$$

$$i^1 = i$$

$$i^2 = \sqrt{-1}^2 = -1$$

$$i^3 = i^2 \times i = -1 \times i$$

$$\Rightarrow i^3 = -i$$

$$\text{Also } i^4 = i^2 \times i^2$$

$$\Rightarrow i^4 = -1 \times -1$$

$$\Rightarrow i^4 = 1$$

Therefore $z^{4R} = (z^4)^R = (1)^R = 1$

$$z^{4R+1} = z^{4R} \times z^1 = z$$

$$z^{4R+2} = z^{4R} \times z^2 = 1 \times -1 = -1$$

$$z^{4R+3} = z^{4R} \times z^3 = 1 \times -i = -i$$

$$\Rightarrow z^{4R} = 1, z^{4R+1} = z, z^{4R+2} = -1, z^{4R+3} = -i$$

Q) Evaluate the following:

(a) z^{25} (b) z^{-31}

(Sol) $z^{25} = z^{4 \times 6 + 1} = z^1 = z$

$$\Rightarrow z^{25} = z$$

$z^{-31} = z^{4 \times -8 + 1}$
 $\Rightarrow z^{-31} = z$

Also, $z^{-31} = \frac{1}{z^{31}} = \frac{1}{z^{4 \times 7 + 3}} = \frac{1}{z^3} = \frac{1}{-i}$

$$\Rightarrow z^{-31} = -\frac{1}{i} \times \frac{i}{i} = \frac{-i}{i^2} = \frac{-i}{+1}$$

$$\Rightarrow z^{-31} = -i$$

QUESTION

The value of i^{100} is

☒ 1 -1

☒ 2 1

☒ 3 i

☒ 4 $-i$

$$i^{100} = i^{4 \times 25} = 1.$$

$$\Rightarrow i^{100} = 1.$$

QUESTION

The value of $i^{30} + i^{40} + i^{60}$ is:

- ☒ 0
- ☒ 1
- ☐ i
- ☐ -1

$$\begin{aligned}
 \text{(Sol)} \quad i^{30} + i^{40} + i^{60} \\
 &= i^2 + i^0 + i^0 \\
 &= -1 + 1 + 1 \\
 &= 1
 \end{aligned}$$

$$\begin{aligned}
 i^{30} &= i^{4 \times 7 + 2} = -1 \\
 i^{40} &= i^{4 \times 10} = 1 \\
 i^{60} &= i^{4 \times 15} = 1
 \end{aligned}$$

QUESTION

The value of i^{2026} is :

- ☒ 1 $-i$
- ☒ 2 1
- ☒ 3 i
- ☒ 4 -1

(Sol) $i^{2026} = i^2$

$\Rightarrow i^{2026} = -1.$

$$2026 = (4 \times 506 + 2) \\ \qquad \qquad \qquad = -1.$$

$$4 \overline{) 2026} \\ \underline{200} \\ 26 \\ \underline{24} \\ 2$$

QUESTION

Which of the following is true?

$\sqrt{ab} = \sqrt{a} \times \sqrt{b}$ if at least one of a & b is positive.

~~1~~ $\sqrt{-2 \times -3} = \sqrt{-2} \times \sqrt{-3}$ ✗

~~2~~ $\sqrt{-2 \times -3} = \sqrt{-2} \times \sqrt{3}$ ✗
 $\sqrt{6} \quad \sqrt{6}i$

~~3~~ $\sqrt{-2 \times -3} = \sqrt{6}$ ✓

~~4~~ $\sqrt{-2 \times -3} = \sqrt{6}i$
 $\ominus \sqrt{6} \neq \sqrt{6}i$ ✗

(Sol) (a) LHS = $\sqrt{-2 \times -3} = \sqrt{6}$

RHS = $\sqrt{-2} \times \sqrt{-3}$

$\Rightarrow$ RHS = $\sqrt{2}i \times \sqrt{3}i$

$\Rightarrow$ RHS = $\sqrt{6}i^2 = \sqrt{6} \times -1$

$\Rightarrow$ RHS = $-\sqrt{6}$

$\sqrt{ab} \neq \sqrt{a} \times \sqrt{b}$ ✗

Complex Number

Definition: A number of the form $a+ib$

where $a, b \in \mathbb{R}$ & $i = \sqrt{-1}$ is known as

a Complex number

* Generally denoted by z . So $z = a+ib$.

* $\text{Re}(z) = a \longrightarrow$ Real part of z .

* $\text{Im}(z) = b \longrightarrow$ Imaginary part of z .

* $2+3i$, $6i = 0+6i$, $3 = 3+2.0$

are all Complex numbers.

* $\text{Re}(z) = 0 \longrightarrow z$ is purely imaginary.

* $\text{Im}(z) = 0 \longrightarrow z$ is purely Real.

* All Real numbers are Complex numbers

* All Imaginary numbers are Complex numbers.

QUESTION

If $z = 6 + 3i$ then the **real and imaginary part of z** are respectively equal to :

1 3 and 6 ✗

2 6 and 3 ✓✓

3 6 and 6 ✗

4 3 and 3 ✗

QUESTION

Let $x, y \in \mathbb{R}$, then $x + iy$ is a non real complex number if:

$$9 = 9 + i0$$

1 $x = 0$ ✗

2 $y = 0$ ✗

3 $x \neq 0$ ✗

4 $y \neq 0$ ✓

QUESTION

Which of the following statement is true?

- 1 $1 + 2i$ is less than $2 + i$ ✗
- 2 Every complex number is a Real number ✗
- 3 The set of complex numbers is not ordered ✓✓
- ~~4~~ None of the above

Thank
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